

Group Topic: *Predicting Student Dropout Risk Using Academic and Behavioral Data*

Lab No.

Git Repo/ Colab Link:

Date:

<https://colab.research.google.com/drive/1ug82XRryCVJDoqn2HZbPz7GTEq-n1545?usp=sharing>

Objectives:

In this task, our goal is to analyze the performance of various machine learning techniques employed for forecasting the chances of student dropout from educational programs by considering relevant data. This activity requires us to preprocess the data set, train various classification models, compare the performance of each model, and find the feature importance to establish the causes of student dropouts.

In particular, the task is geared towards:

1. Pre-processing the dataset for use in machine learning
2. Training and validating several classification models
3. Comparing the accuracy and predictive capabilities of each model
4. Tuning the hyperparameters of each model
5. Analysing the results from the feature importance analysis
6. Enhancing one's comprehension of model evaluation methodologies

Tasks Completed:

The following tasks were completed as part of the ***Model Evaluation & Visualization role:***

1. Evaluated Multiple Machine Learning Models:

Tested and compared different classification models to determine which model performs best in predicting student dropout risk.

Models Evaluated:

- Logistic Regression
- Random Forest Classifier
- XGBoost Classifier

Code Used:

```
models = {  
    'Logistic Regression': LogisticRegression(random_state=42, max_iter=1000),  
    'Random Forest': RandomForestClassifier(random_state=42),  
    'XGBoost': XGBClassifier(random_state=42, eval_metric='logloss', use_label_encoder=False)  
}
```

2. Performed Model Training and Accuracy Evaluation

Trained each model using the training dataset and evaluated prediction performance using accuracy scores.

Process Performed:

- Trained models using `model.fit()`
- Generated predictions using `model.predict()`
- Compared prediction accuracy of each classifier

Code Used:

```
results = {}  
for name, model in models.items():  
    model.fit(X_train, y_train)  
    y_pred = model.predict(X_test)  
    accuracy = accuracy_score(y_test, y_pred)  
    results[name] = {'model': model, 'accuracy': accuracy, 'predictions': y_pred}  
    print(f'{name} Accuracy: {accuracy:.4f}')
```

3. Conducted Hyperparameter Tuning on XGBoost

Optimized the XGBoost classifier using `GridSearchCV` to improve prediction performance.

Parameters Tuned:

- `n_estimators`
- `max_depth`
- `learning_rate`
- `subsample`

Code Used:

```
from sklearn.model_selection import GridSearchCV

param_grid = {
    'n_estimators': [100, 200],
    'max_depth': [3, 5, 7],
    'learning_rate': [0.01, 0.1, 0.2],
    'subsample': [0.8, 1.0]
}

xgb = XGBClassifier(random_state=42, eval_metric='logloss')
grid = GridSearchCV(xgb, param_grid, cv=5, scoring='roc_auc', n_jobs=-1)
grid.fit(X_train, y_train)
```

Results and Analysis

The machine learning models were successfully analyzed using the student academic and behavioral data set. The comparison was done for three models – Logistic Regression, Random Forest Classifier and XGBoost model.

During the testing process, it was found out that the XGBoost model has shown the highest performance level after the hyperparameter tuning using GridSearchCV. Hyperparameter tuning helped in enhancing the prediction capability of the XGBoost model by choosing optimal parameters through ROC-AUC scoring.

Accuracy scores obtained during testing allowed estimating the effectiveness of each model in predicting the risk of student dropout. While Logistic Regression model offered quite stable results, Random Forest improved the model performance; however, it was found out that XGBoost is the most effective prediction tool.

Besides, the analysis of feature importance made it possible to identify the variables, which played the most important role in predicting the student dropout. This allowed the group to identify the main academic and behavioral aspects affecting the student dropout.

All things considered, it is possible to state that testing and analysis of models have proved their usefulness.

Challenges and Solutions:

Challenge:

Some machine learning models produced lower prediction accuracy during initial testing.

Solution:

Compared multiple classification algorithms and selected the best-performing model based on evaluation metrics.

Challenge:

The XGBoost model required additional optimization for better performance.

Solution:

Used `GridSearchCV` to perform hyperparameter tuning and improve model accuracy.

Challenge:

Difficulty identifying which features most influenced student dropout prediction.

Solution:

Used feature importance analysis from the optimized XGBoost model to rank the most significant variables.

Conclusion:

The goals of the activity were met successfully by optimizing machine learning algorithms to predict student dropout.

This activity enhanced my knowledge about model evaluation, hyperparameter tuning, and feature selection with Python, Scikit-learn, and XGBoost. In terms of the classification models explored, the XGBoost algorithm gave the best prediction performance after tuning.

From this activity, it is evident that comparing and evaluating models is crucial in building efficient predictive systems. Moreover, feature importance analysis was useful in analyzing student dropouts' influencing factors.